AI Assistant: Pharyngitis Classification (Bacterial vs. Viral)

Group_A1_85
José Torres
Rodrigo Soares
Vicente Silva

U.PORTO
FEUP FACULDADE DE ENGENHARIA
UNIVERSIDADE DO PORTO

Customer

U.PORTO
FEUP FACULDADE DE ENGENHARIA
UNIVERSIDADE DO PORTO

- **Customer:** USF Nova Era / ULS Tâmega e Sousa
- **Contact Person:** Maria Luísa Ferreira, Family Doctor

Problem

U. PORTO
FEUP FACULDADE DE ENGENHARIA UNIVERSIDADE DO PORTO

- **Definition of the problem:** It is clinically difficult to quickly and accurately distinguish between Streptococcal Pharyngitis (bacterial) and common viral throat infections based solely on visual observation.

- **Current approach:** Doctors rely on subjective clinical scoring (e.g., Centor criteria) or must wait for throat swab cultures and rapid antigen tests, which delays the correct treatment or leads to prescribing antibiotics "just in case".

Available data

U.PORTO
FEUP FACULDADE DE ENGENHARIA
UNIVERSIDADE DO PORTO

- **High-level description:** A robust dataset of 50000 synthetic patient records representing clinical triage data.

- **Entities/Tables:** Patient demographics (Age)
  - Clinical Symptoms (Fever, Cough, Tonsil Exudate, Lymph Nodes, etc);
  - Final Diagnosis (The ground truth established by hypothetical lab tests).

Solution overview

U.PORTO
FEUP FACULDADE DE ENGENHARIA
UNIVERSIDADE DO PORTO

- **Description of the solution:** A Machine Learning-based clinical decision support system designed to assist doctors in real-time triage.

- **Other components besides ML:** A responsive Web Application built with Streamlit, where doctors can input symptoms and instantly receive an infection probability score and treatment recommendations.

Predictive model

- **Target Variable:** Infection Type (1 = Bacterial/Streptococcus, 0 = Viral).

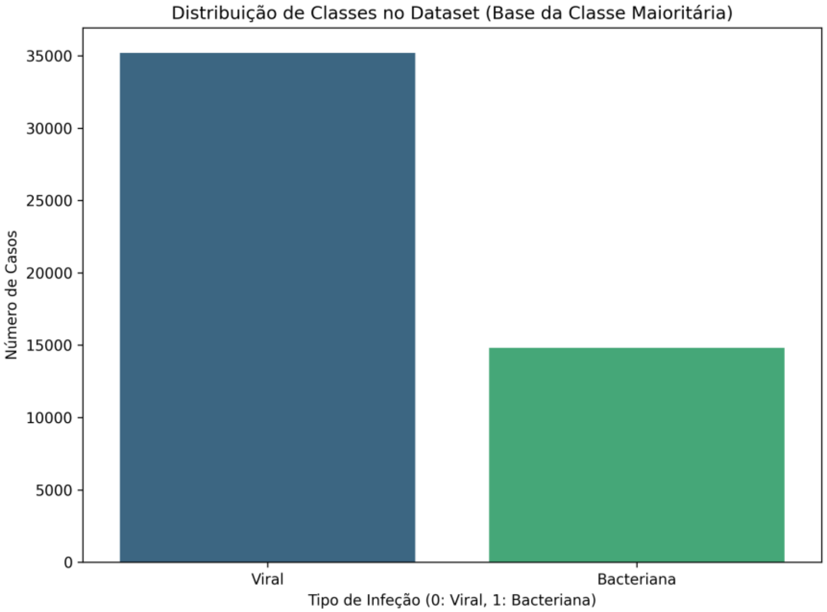
- **Attributes (Features):** Age, Sudden Onset, Body Temperature, Throat Pain Intensity, Symptom Duration, Tonsil Hypertrophy, Exudate Level, Swollen Lymph Nodes, Abdominal Pain, Cough Type, Coryza, Conjunctivitis, Palatal Petechiae, Myalgia, Headache, Halitosis, and Painful Adenopathy.

Empirical study: setup

- **Artificial data generated:** 50,000 patient records were generated using probability distributions based on real medical criteria (Centor/McIsaac scores), including 2% of random noise to simulate human error and atypical cases.

- **Model evaluation process:** The dataset was split into 70% for training and 30% for testing.

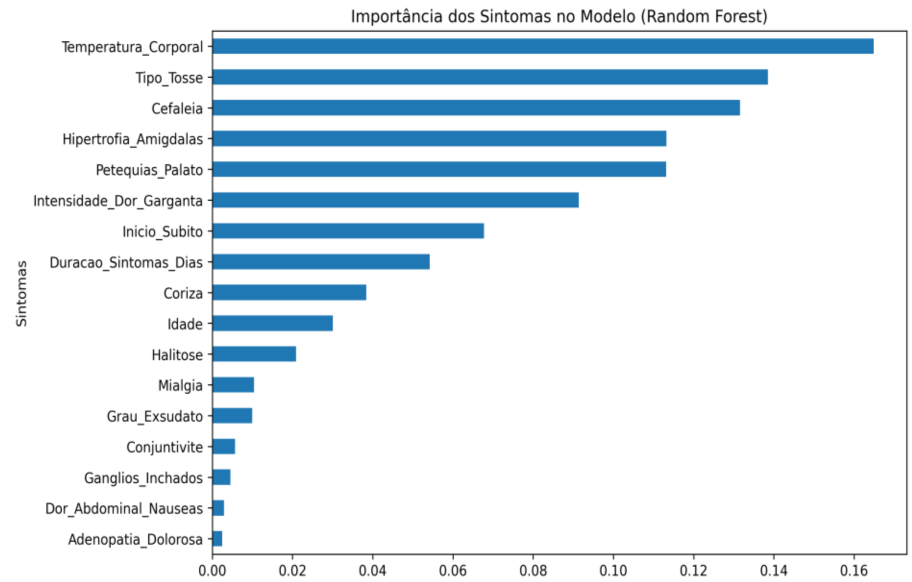
- **Evaluation measures:** Accuracy, Precision, Recall, and F1-Score, with a special focus on Recall to minimize false negatives in medical diagnosis.

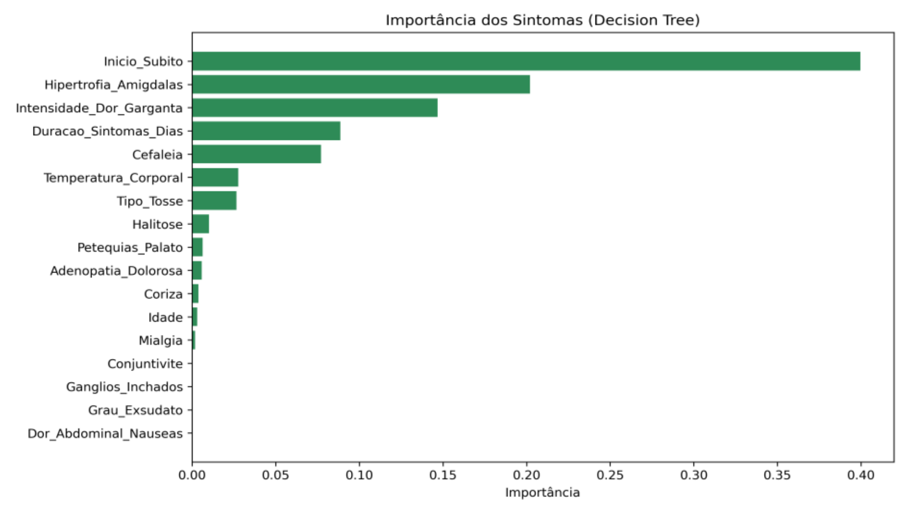


Empirical study: results

U.PORTO
FEUP FACULDADE DE ENGENHARIA
UNIVERSIDADE DO PORTO

- **Random Forest Classifier:** aproximately 94.6% Accuracy. Provided the highest stability and predictive power.
- **Decision Tree Classifier:** aproximately 93.6% Accuracy. Slightly less accurate but provides highly valuable visual explainability for doctors.
- **Majority Class (Baseline):** aproximately 70.4% Accuracy. Proves that our ML models learned complex symptom relationships beyond just guessing the most common viral infection.





Stakeholders

U.PORTO
FEUP FACULDADE DE ENGENHARIA
UNIVERSIDADE DO PORTO

- **Medical Doctors:** The primary end-users who will use the app to support their clinical decisions.

- **Patients:** Beneficiaries of faster diagnoses and more appropriate treatments.

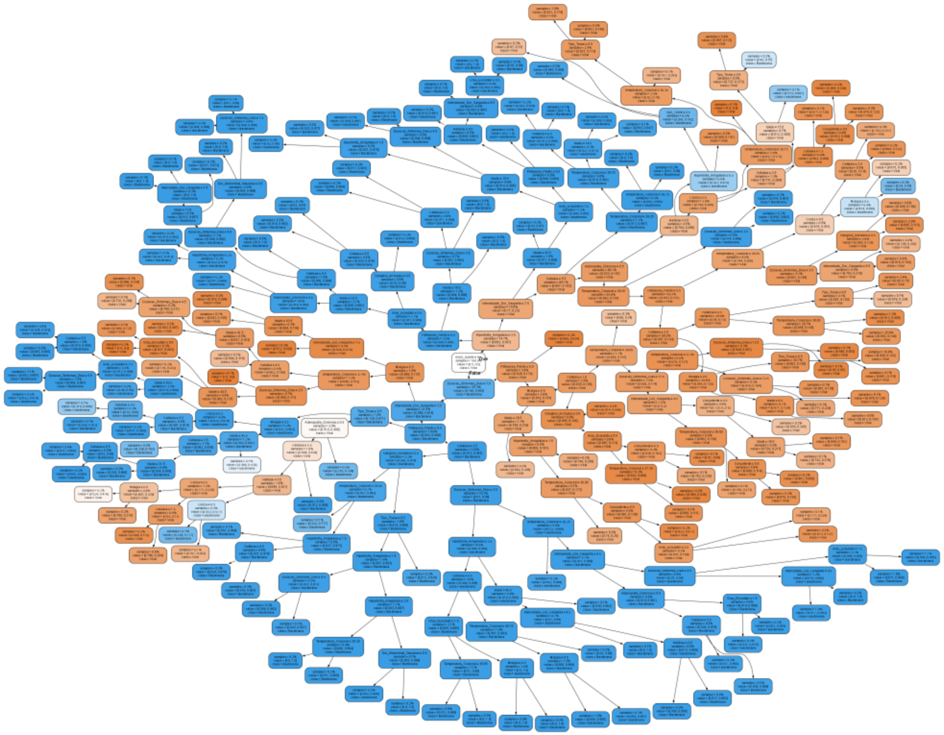
- **Hospital / Clinic Administration:** Interested in operational efficiency and reducing costs from unnecessary lab tests.

- **IT / ML Developers:** Responsible for maintaining the model and keeping the web application secure.

Expected challenges during development and deployment

- **Technical:** Ensuring the generated artificial data perfectly mimics the complexity of the real world; handling missing or subjectively reported symptoms by patients.

- **Organizational:** Overcoming the natural skepticism and resistance from medical professionals who might not initially trust an Artificial Intelligence over their own clinical experience.



Unavailable data

U.PORTO
FEUP FACULDADE DE ENGENHARIA
UNIVERSIDADE DO PORTO

- **Additional data that should be available:** Historical patient medical records (comorbidities, previous antibiotic usage), real rapid strep test results, and high-resolution images of the patients' tonsils.

- **Challenges involved in collecting it:** Strict medical privacy laws (GDPR/HIPAA) make it very hard to acquire real patient data. Additionally, securely collecting and annotating medical images requires a huge and expensive effort from healthcare professionals.

Ethical risks

U.PORTO FEUP FACULDADE DE ENGENHARIA UNIVERSIDADE DO PORTO

- **False Negatives (Missed Bacteria):** The model predicts a viral infection, the patient is sent home without antibiotics, and the bacterial infection worsens, potentially leading to severe complications like rheumatic fever.

- **False Positives (Over-prescription):** The model incorrectly predicts a bacterial infection, leading the doctor to prescribe unnecessary antibiotics. This directly contributes to the global crisis of antimicrobial resistance.

- **Mitigation:** The system must be explicitly labeled as a "Decision Support Tool", ensuring the final diagnosis always remains the doctor's responsibility.